

John Evans Dakurah

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EDUCATION

Ankara Yıldırım Beyazıt University | *MSc in Mechanical Engineering*

Ankara, Turkey | Sept, 2025 – June, 2027

- CGPA: **4.00 / 4.00**

Gazi University | *BSc in Automotive Engineering*

Ankara, Turkey | Sept, 2021 – June, 2025

- CGPA: **3.56 / 4.00**
- Class Rank: **1 / 115**

HONORS & AWARDS

- **Best Graduating Student, Department of Automotive Engineering** — Gazi University, 2025

RESEARCH EXPERIENCE

Graduate Research Assistant | *Prof. Kamil Arslan*

Ankara, Turkey | Sept, 2025 – Present

- Conduct computational and experimental research in thermal and fluid sciences to investigate fluid flow and heat transfer in energy systems.
- Develop and validate numerical models using mesh-independence studies, parametric analyses, experimental data, and numerical post-processing to evaluate system performance under different operating conditions.
- Contribute to research on thermal-fluid systems and energy-conversion technologies, with applications in fuel cells, internal combustion engines, heat exchangers, and thermal management.
- Collaborate with researchers in biomedical engineering on computational and experimental studies of cyclotide transport in membrane-based microfluidic systems, examining mass-transfer behavior under varying pH and flow-rate conditions.

Undergraduate Research Assistant | *Prof. Hamit Solmaz*

Ankara, Turkey | Sept, 2022 – Aug, 2024

- Developed physics-based models and simulations of lithium-ion battery electric, fuel-cell electric, and hybrid electric vehicles using MATLAB and Simulink.
- Developed energy-management systems for hybrid electric vehicles using fuzzy-logic and conventional on-off control, emphasizing energy efficiency and optimization.
- Developed electrochemical models of proton-exchange-membrane (PEM) fuel cells for electric-vehicle applications.

JOURNAL PUBLICATIONS

- Dakurah, J. E., & Arslan, K. (2026). Numerical investigation of nanofluid coolant and cooling channel geometry for thermal management in PEM fuel cells. *Engineering Perspective* (Under Review).
- Dakurah, J. E., Solmaz, H., & Kocakulak, T. (2024). Modeling of a PEM Fuel Cell Electric Bus with MATLAB/Simulink. *Automotive Experiences*, 7(2), 252-269. <https://doi.org/10.31603/ae.11471>
- Dakurah, J. E. (2026). Fuel Cell Electric Vehicles: A Review of Technology, Energy Management Strategies, Challenges, and Future Research Directions. *International Journal of Automotive Engineering and Technologies*. (Under Review).

TECHNICAL SKILLS

- **Simulation & Modeling:** ANSYS Fluent, ANSYS Workbench (SpaceClaim, DesignModeler), MATLAB, Simulink
- **Programming & Data Analysis:** Python, FORTRAN, NumPy, Pandas, Matplotlib, Seaborn, Jupyter Notebook, OriginPro
- **Research Methods:** Computational fluid dynamics (CFD), physics-based numerical modeling, mesh-independence studies, parametric analysis, experimental validation, numerical post-processing, data visualization
- **CAD & Engineering Design:** SolidWorks, CATIA V5
- **Additional Computing:** Linear and logistic regression, decision trees, support vector machines, random forests, JavaScript, HTML, CSS, React, Redux, Bootstrap, jQuery, SASS
- **UAV & Flight-Control Tools:** PyMavlink, MAVProxy, Mission Planner

WORK EXPERIENCE

Automotive Engineering Intern | *Barok Defense*

İzmir, Turkey | July, 2024 – Jan, 2025

- Performed computational thermal and fluid-flow analyses using ANSYS to evaluate heat-transfer and flow behavior in camera, lighting, communication, and fuel-heating systems for military vehicles.
- Compared alternative system designs, identified promising configurations, and supported design refinement based on simulation results and engineering requirements.
- Built and tested prototypes through experimental thermal, fluid, mechanical, and functional validation to identify performance issues and verify system reliability under demanding operating conditions.

TEACHING EXPERIENCE

Science and Mathematics Tutor | *Portakal Eğitim Kurumu*

Ankara, Turkey | Feb, 2025 – Present

- Teach mathematics and science subjects, including physics, chemistry, and biology, to high school students through clear, structured instruction.
- Prepare students for high school examinations using targeted review, practice problems, and progress-based academic support.
- Provide individual guidance and mentoring to help students strengthen subject understanding, problem-solving skills, and study habits.

ACADEMIC PROJECTS

- Comparative CFD Analysis of Exhaust Manifold Designs** | *ANSYS Fluent* **Feb, 2026 – May, 2026**
- Compared collector-type and log-type exhaust manifold configurations using computational fluid dynamics to assess flow distribution and thermal behavior.
 - Evaluated velocity, pressure, turbulence, and temperature characteristics to identify design effects on exhaust-flow performance.
- Four-Cylinder Diesel Engine Design & Analysis** | *SolidWorks, ANSYS Workbench* **May, 2024 – June, 2024**
- Designed a four-cylinder, water-cooled diesel engine using SolidWorks.
 - Performed steady-state thermal analysis of the engine using ANSYS Workbench.
- Flight Control Programming of a UAV** | *Python, PyMAVLink, Mission Planner* **Jan, 2024 – March, 2024**
- Implemented MAVLink communication between a UAV and a ground station using Python and PyMAVLink.
 - Implemented MAVLink protocols to send and receive flight information from the UAV.
 - Implemented MAVLink protocols to control UAV movement, including speed, yaw, pitch, roll, and landing.
- Design & Analysis of an M3 Category Articulated Bus** | *SolidWorks, CATIA V5* **Sept, 2023 – Dec, 2023**
- Developed a 3D model of a 17.1 meter long articulated bus using CATIA V5
 - Performed structural analyses of the model using SolidWorks.